8.2.7 Stability

During engine's functioning, the lubricating oil should reduce the friction without any oxidation. If lubricating oil is oxidised, it produces acids which leads to dirt and corrosion of the engine parts.

8.2.8 Corrosiveness

During engine's functioning, several parts of the engine are subjected to corrosion due to the chemicals present in the lubricating oil. Hence, lubricating oils should have very less amount of acids and chemicals which will cause corrosion. This is called corrosiveness.

8.2.9 Cleanliness

Lubricating oils should possess the property of cleanliness so that dusts and unwanted materials in the lubricant could be removed. This property helps in cleaning the carbon deposits in the lubricant due to burning of fuels. This is called cleanliness. Normally inorganic lubricants have this property more than organic lubricants.

8.2.10 Emulsification

If lubricating oil is dissolved in water, it results in emulsification. If it dissolves in water, it will lose the property of lubrication. To prevent this additives are added along with the lubricating oil.

8.2.11 Cloud Point

The Temperature at which lubricating oil first changes its phase from liquid state to solid state is known as cloud point.

8.2.12 Foaming

During lubrication action, lubricating oils will have large amount of very small air bubbles to be present. This process is called foaming. It may lead to oxidation and also these bubbles get deposited on the friction surface and prevent the lubricant to flow through the surfaces.

8.3 TYPES OF LUBRICANTS

The lubricants used in machines and automobiles are listed below:

1) Solid lubricant.

2) Semi solid lubricant.

3) Liquid lubricant.

8.3.1 Solid Lubricant

Lubricant materials available in solid state are called as solid lubricants. The solid substance like fibre, graphite, carbon, mica, vax are some of the examples for solid lubricants. They are used in places where liquid lubricants can't be used and also used in high temperature places. See Figure shown in 8.3.1 Solid Lubricant

8.3.2 Semi Solid Lubricant

Lubricant materials present at the state in-between solid and liquid are called semi solid lubricants. The places at high stress and where liquid and solid lubricants cannot be used, semi-solid lubricants are used. In automobiles, in all the bearings other than engine bearing the semi solid lubricants are used. The following table shows the examples for semi solid lubricants and their applications in automobiles. See Figure shown in 8.3.2, Semi Solid Lubricant.

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